

NATALIA ADAMOWSKA

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Creative and open-minded chemistry graduate, specialising in data analysis.

5th year Master of Chemistry with Placement graduate, with a First-Class degree awarded in June 2025, with experience in analysing and processing big amounts of data efficiently and creatively, proactive and adaptable approach, guided by natural curiosity, analytical and critical thinking to solve problems, developing through collaboration with an international and multidisciplinary teams.

Awarded multiple times for an academic excellence. Participant of academic internships in physical sciences, contributing to the understanding the plasma erosion and adhesion of particles for the high velocity impacts.

SKILLS

Technical

- Programming in Python, MATLAB, Java.
- Data analysis in Python (Pandas, Numpy) and Excel.
- GitHub knowledge and project [portfolio](#).
- Proficient MS Excel, Word and PowerPoint skills.
- Image analysis using ImageJ software.
- Research, chemical and mathematical skills.
- Writing scientific reports, papers and reviews.
- Laboratory skills, SOP writing.

Non-technical

- Critical and analytical thinking.
- Proactive, adaptable and flexible.
- Problem solving and willingness to learn.
- Ambitious and motivated.
- Presentation and communication.
- Organisation and time management.
- Attention to detail.
- Natural curiosity.

PROJECTS

“Anharmonic vibrational spectra for the molecules found in exoplanetary atmospheres.”

Sept 2023 – June 2024 (9mo), University of New South Wales, Sydney.

- Conducting a quantum computational chemistry benchmark study with applications in astronomy.
- Managing and analysing big amounts of data, using Python and Excel, later used to develop innovative AI models.
- Developing proactive and flexible problem-solving approach, showing continuous engagement and curiosity.
- Collaborating with a multidisciplinary and international team to finalise projects before deadlines.
- Participating in weekly group meetings to assess progress and future research directions.
- Using Gaussian and basic bash Linux scripts to calculate vibrational frequencies.
- Critically evaluating literature to write a high-quality scientific report, review and journal paper, using LaTeX.

“Crossed-molecular beam studies of the rotationally correlated inelastic scattering of NO radicals.”

Sept 2024 – June 2025 (9mo), Heriot-Watt University, Edinburgh.

- Studying rotational energy transfers in gas phase inelastic molecular collisions.
- Analysing numerical data and images, using Excel and ImageJ, to find trends and patterns in studied systems.
- Applying critical and analytical thinking to drive project, address complex issues and find fresh perspectives.
- Effectively communicating within an international and multidisciplinary team.
- Using MC Word to write high-quality scientific report and literature review.

EDUCATION

Sept 2020 – June 2025 (MChem) Chemistry with Placement, Heriot-Watt University, Edinburgh.

- Researching chemistry, in synthetic, materials and laser laboratories, and using computational methods.
- Programming in Python, MATLAB, Java.

- Using Python, Excel, MATLAB and ImageJ for data and image analysis, plotting and fitting.
- Using Word and PowerPoint to write reports, design posters and presentations.
- Developing communication, presentation and interpersonal skills by presenting at seminars.

Sept 2023 – June 2024 International Research Placement, University of New South Wales, Sydney.

- Researching astrochemistry using quantum computational methods.
- Developing communication and presentation skills by presenting at talks and conferences, like 2023 RACI Physical Chemistry Student Conference in Katoomba, Australia.
- Participating in an international exchange.
- Collaborating with an international and multidisciplinary team, growing my interpersonal and teamwork skills.

EXPERIENCE

June 2024 – Aug 2024 (2mo), Chemistry Intern, Heriot-Watt University, Edinburgh.

“Adhesion of silica particles to gold film for high velocity impacts.”

- Developing laboratory, research and analytical skills by working with lasers, polymers and coatings.
- Studying material properties to evaluate their performance for high velocity impacts experiments.
- Utilising SEM, Python and Excel for data and image analysis.
- Regularly presenting research findings to a group of international academics.
- Using Overleaf and MC Word to write high-quality scientific reports.
- Using basic CAD and 3D printers to design sample holder.

June 2023 – Aug 2023 (2mo), Chemistry Intern, Heriot-Watt University, Edinburgh.

“Understanding Plasma Erosion of Plastics Waste”

- Using plasma oven and basic gas pumping systems, working in a materials laboratory.
- Working with plastics, such as PVC and HDPE to study alternative recycling methods.
- Using IR spectroscopy and MATLAB to analyse data, and characterise different materials.
- Collaborating with a research group, growing interpersonal skills, participating in weekly meetings.
- Investigate the effect of plasma on nitrile gloves and their dissolution in different solvents.

Mar 2022 – July 2023 (1y 6mo), Heriot-Watt University Ambassador, Heriot-Watt University, Edinburgh.

- Demonstrating chemistry experiments and explaining the principles behind them to prospective students.
- Managing the organisation of Open Days, growing interpersonal and project management skills.

Sept 2021 – May 2025 (3y 6mo) ELAB Ambassador, ELAB Education Laboratory.

- Sharing my experience of studying abroad, developing communication and interpersonal abilities.
- Created and hosted study-themed content on Facebook, Instagram and TikTok, developing presentation, design, creativity and attention to detail skills.
- Writing essays on studying abroad, taking part in interviews and recording podcasts.

AWARDS

AWE Prize for academic year 2024/2025

For outstanding performance by in either MChem in Chemistry or MChem with Industrial Experience.

Deputy Principal's Award for academic years 2022/2023, 2021/2022 and 2020/2021

For academic excellence, achieved Grade A, or equivalent, in six or more courses.

Nisbet Memorial Prize for academic year 2022/2023

Awarded to the most distinguished student in the programme for the degree of BSc or MChem in Chemistry.

HOBBIES

Skiing, bouldering and photography.